line to the guide. Conversely, the center conductor can act like a probe, coupling a signal from the waveguide to the coaxial cable.

For guided transmission at frequencies below 30 GHz, the coaxial cable is by far the most widely used transmission line. At higher frequencies, however, the coaxial cable has a number of limitations: (a) in order for it to propagate only TEM modes, the cable's inner and outer conductors have to be reduced in size to satisfy a certain size-to-wavelength requirement, making it more difficult to fabricate, (b) the smaller cross section reduces the cable's power-handling capacity (limited by dielectric breakdown), and (c) the attenuation due to dielectric losses increases with frequency. For all of these reasons, metal waveguides have been used as an alternative to coaxial lines for many radar and communication applications that operate at frequencies in the 5–100 GHz range, particularly those requiring the transmission of high levels of radio-frequency (RF) power. Even though waveguides with circular and elliptical cross sections have been used in some microwave systems, the rectangular shape has been the more prevalent geometry.

8-7 General Relations for **E** and **H**

The purpose of the next two sections is to derive expressions for **E** and **H** for the TE and TM modes in a rectangular waveguide and to examine their wave properties. We choose the coordinate system shown in **Fig. 8-22**, where the propagation occurs along $\hat{\mathbf{z}}$. For TE modes, the electric field is transverse to the direction of propagation. Hence, **E** may have components along $\hat{\mathbf{x}}$ and $\hat{\mathbf{y}}$ but not along $\hat{\mathbf{z}}$. In contrast, **H** has a $\hat{\mathbf{z}}$-directed component and may have components along either $\hat{\mathbf{x}}$ or $\hat{\mathbf{y}}$ or along both. The converse is true for TM modes.

Our solution procedure consists of four steps:

(1) Maxwell's equations are manipulated to develop general expressions for the phasor-domain transverse field components $\widetilde{E}_x$, $\widetilde{E}_y$, $\widetilde{H}_x$, and $\widetilde{H}_y$ in terms of $\widetilde{E}_z$ and $\widetilde{H}_z$. When specialized to the TE case, these expressions become functions of $\widetilde{H}_z$ only, and the converse is true for the TM case.

(2) The homogeneous wave equations given by Eqs. (7.15) and (7.16) are solved to obtain valid solutions for $\widetilde{E}_z$ (TM case) and $\widetilde{H}_z$ (TE case) in a waveguide.

(3) The expressions derived in step 1 are then used to find $\widetilde{E}_x$, $\widetilde{E}_y$, $\widetilde{H}_x$, and $\widetilde{H}_y$.

(4) The solutions obtained in step 3 are analyzed to determine the phase velocity and other properties of the TE and TM waves.

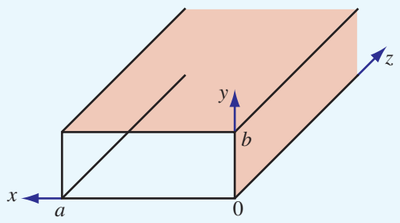


Figure 8-22 Waveguide coordinate system.

The intent of the present section is to realize the stated goals of step 1. We begin with a general form for the **E** and **H** fields in the phasor domain:

$$\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\widetilde{E}_x + \hat{\mathbf{y}}\widetilde{E}_y + \hat{\mathbf{z}}\widetilde{E}_z, \tag{8.84a}$$

$$\widetilde{\mathbf{H}} = \hat{\mathbf{x}}\widetilde{H}_x + \hat{\mathbf{y}}\widetilde{H}_y + \hat{\mathbf{z}}\widetilde{H}_z. \tag{8.84b}$$

In general, all six components of $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ may depend on (x,y,z), and while we do not yet know how they functionally depend on (x,y), our prior experience suggests that $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ of a wave traveling along the $+z$ direction should exhibit a dependence on z of the form $e^{-j\beta z}$, where β is a yet-to-be determined ***phase constant***. Hence, we adopt the form

$$\widetilde{E}_x(x,y,z) = \widetilde{e}_x(x,y)\, e^{-j\beta z}, \tag{8.85}$$

where $\widetilde{e}_x(x,y)$ describes the dependence of $\widetilde{E}_x(x,y,z)$ on (x,y) only. The form of Eq. (8.85) can be used for all other components of $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ as well. Thus,

$$\widetilde{\mathbf{E}} = (\hat{\mathbf{x}}\widetilde{e}_x + \hat{\mathbf{y}}\widetilde{e}_y + \hat{\mathbf{z}}\widetilde{e}_z)e^{-j\beta z}, \tag{8.86a}$$

$$\widetilde{\mathbf{H}} = (\hat{\mathbf{x}}\widetilde{h}_x + \hat{\mathbf{y}}\widetilde{h}_y + \hat{\mathbf{z}}\widetilde{h}_z)e^{-j\beta z}. \tag{8.86b}$$

The notation is intended to clarify that, in contrast to $\widetilde{E}$ and $\widetilde{H}$, which vary with (x,y,z), the lower case $\widetilde{e}$ and $\widetilde{h}$ vary with (x,y) only.

In a lossless, source-free medium (such as the inside of a waveguide) characterized by permittivity ε and permeability μ (and conductivity $\sigma = 0$), Maxwell's curl equations are given by Eqs. (7.2b and d) with $\mathbf{J} = 0$,

$$\nabla \times \widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}}, \tag{8.87a}$$

$$\nabla \times \widetilde{\mathbf{H}} = j\omega\varepsilon\widetilde{\mathbf{E}}. \tag{8.87b}$$

Upon inserting Eqs. (8.86a and b) into Eqs. (8.87a and b) and recalling that each of the curl equations actually consists of

线到指南。相反，中心导体可以像探针一样，将信号从波导耦合到同轴电缆。

对于 30 GHz 以下频率的引导传输，同轴电缆是迄今为止使用最广泛的传输线。然而，在较高频率下，同轴电缆有许多限制：(a) 为了使其仅传播 TEM 模式，必须减小电缆的内导体和外导体的尺寸以满足一定的尺寸与波长要求，从而使其更难以制造，(b) 较小的横截面会降低电缆的功率处理能力（受到介电击穿的限制），以及 (c) 由于介电损耗导致的衰减随着频率的增加而增加。由于所有这些原因，金属波导已被用作许多雷达和通信应用中同轴线的替代品，这些雷达和通信应用的工作频率在 5-100 GHz 范围内，特别是那些需要传输高水平射频 (RF) 功率的应用。尽管具有圆形和椭圆形横截面的波导已经在一些微波系统中使用，但矩形形状是更普遍的几何形状。

8-7 E和H的一般关系

接下来两节的目的是推导矩形波导中 TE 和 TM 模式的 E 和 H 表达式，并检查它们的波特性。我们选择图 8-22 所示的坐标系，其中传播沿 $\hat{\mathrm{z}}$ 发生。对于 TE 模式，电场垂直于传播方向。因此，E 可能具有沿 $\hat{\mathrm{x}}$ 和 $\hat{\mathrm{y}}$ 的分量，但不沿 $\hat{\mathrm{z}}$ 的分量。相反，H 具有 $\hat{\mathrm{z}}$ 方向的分量，并且可以具有沿 $\hat{\mathrm{x}}$ 或 $\hat{\mathrm{y}}$ 或沿两者的分量。对于 TM 模式则相反。

我们的解决方案过程包括四个步骤：

(1) 操纵麦克斯韦方程，以 $\widetilde{\mathrm{E}}_z$ 和 $\widetilde{\mathrm{H}}_z$ 的形式，建立相量域横向场分量 $\widetilde{\mathrm{E}}_x$、$\widetilde{\mathrm{E}}_y$、$\widetilde{\mathrm{H}}_x$ 和 $\widetilde{\mathrm{H}}_y$ 的一般表达式。什么时候

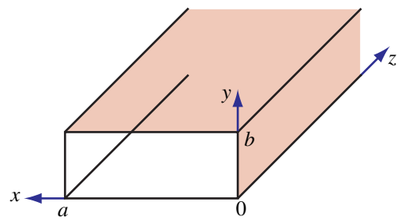


图 8-22 波导坐标系。

专门针对 TE 情况，这些表达式仅成为 $\widetilde{\mathrm{H}}_z$ 的函数，而对于 TM 情况则相反。

(2) 齐次波动方程由方程给出。求解（7.15）和（7.16）以获得波导中 $\widetilde{\mathrm{E}}_z$ (TM情况)和 $\widetilde{\mathrm{H}}_z$ (TE情况)的有效解。

(3) 然后使用步骤 1 中导出的表达式来查找 $\widetilde{\mathrm{E}}_x$、$\widetilde{\mathrm{E}}_y$、$\widetilde{\mathrm{H}}_x$ 和 $\widetilde{\mathrm{H}}_y$。

(4) 对步骤3中得到的解进行分析，以确定TE波和TM波的相速度和其他特性。

本节的目的是实现步骤 1 的既定目标。我们从相量域中 E 和 H 场的一般形式开始：

$$\widetilde{\mathbf{E}} = \hat{\mathbf{x}}\widetilde{E}_x + \hat{\mathbf{y}}\widetilde{E}_y + \hat{\mathbf{z}}\widetilde{E}_z, \tag{8.84a}$$

$$\widetilde{\mathbf{H}} = \hat{\mathbf{x}}\widetilde{H}_x + \hat{\mathbf{y}}\widetilde{H}_y + \hat{\mathbf{z}}\widetilde{H}_z. \tag{8.84b}$$

一般来说，$\widetilde{\mathrm{E}}$ 和 $\widetilde{\mathrm{H}}$ 的所有六个分量都可能依赖于 (x, y, z)，虽然我们还不知道它们在功能上如何依赖于 (x, y)，但我们之前的经验表明，沿着 +z 方向传播的波的 $\widetilde{\mathrm{E}}$ 和 $\widetilde{\mathrm{H}}$ 应该表现出对 z 形式 $\mathrm{e}^{-j\beta z}$ 的依赖性，其中 β 是尚未确定的相位常数。因此，我们采用以下形式

$$\widetilde{E}_x(x,y,z) = \widetilde{e}_x(x,y)\, e^{-j\beta z}, \tag{8.85}$$

其中 $\widetilde{\mathrm{e}}_x(x, y)$ 描述了 $\widetilde{\mathrm{E}}_x(x, y, z)$ 仅对 (x, y) 的依赖性。方程的形式(8.85) 也可用于 $\widetilde{\mathrm{E}}$ 和 $\widetilde{\mathrm{H}}$ 的所有其他分量。因此，

$$\widetilde{\mathbf{E}} = (\hat{\mathbf{x}}\widetilde{e}_x + \hat{\mathbf{y}}\widetilde{e}_y + \hat{\mathbf{z}}\widetilde{e}_z)e^{-j\beta z}, \tag{8.86a}$$

$$\widetilde{\mathbf{H}} = (\hat{\mathbf{x}}\widetilde{h}_x + \hat{\mathbf{y}}\widetilde{h}_y + \hat{\mathbf{z}}\widetilde{h}_z)e^{-j\beta z}. \tag{8.86b}$$

该符号旨在澄清，与随 (x, y, z) 变化的 $\widetilde{\mathrm{E}}$ 和 $\widetilde{\mathrm{H}}$ 不同，小写的 $\widetilde{\mathrm{e}}$ 和 $\widetilde{\mathrm{h}}$ 仅随 (x, y) 变化。

在以介电常数 ε 和磁导率 μ（以及电导率 $\sigma = 0$) 为特征的无损、无源介质（例如波导内部）中，麦克斯韦旋度方程由等式 1 给出。 (7.2b 和 d) 其中 $\mathrm{J} = 0$，

$$\nabla \times \widetilde{\mathbf{E}} = -j\omega\mu\widetilde{\mathbf{H}}, \tag{8.87a}$$

$$\nabla \times \widetilde{\mathbf{H}} = j\omega\varepsilon\widetilde{\mathbf{E}}. \tag{8.87b}$$

插入方程式后。 (8.86a 和 b）代入等式。 (8.87a 和 b）并回顾每个旋度方程实际上由

three separate equations—one for each of the unit vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$, we obtain the following relationships:

$$\frac{\partial \widetilde{e}_z}{\partial y} + j\beta \widetilde{e}_y = -j\omega\mu \widetilde{h}_x, \tag{8.88a}$$

$$-j\beta \widetilde{e}_x - \frac{\partial \widetilde{e}_z}{\partial x} = -j\omega\mu \widetilde{h}_y, \tag{8.88b}$$

$$\frac{\partial \widetilde{e}_y}{\partial x} - \frac{\partial \widetilde{e}_x}{\partial y} = -j\omega\mu \widetilde{h}_z, \tag{8.88c}$$

$$\frac{\partial \widetilde{h}_z}{\partial y} + j\beta \widetilde{h}_y = j\omega\varepsilon \widetilde{e}_x, \tag{8.88d}$$

$$-j\beta \widetilde{h}_x - \frac{\partial \widetilde{h}_z}{\partial x} = j\omega\varepsilon \widetilde{e}_y, \tag{8.88e}$$

$$\frac{\partial \widetilde{h}_y}{\partial x} - \frac{\partial \widetilde{h}_x}{\partial y} = j\omega\varepsilon \widetilde{e}_z. \tag{8.88f}$$

Equations (8.88a-f) incorporate the fact that differentiation with respect to z is equivalent to multiplication by $-j\beta$. By manipulating these equations algebraically, we can obtain expressions for the x and y components of $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ in terms of their z components, namely

$$\widetilde{E}_x = \frac{-j}{k_c^2}\left(\beta \frac{\partial \widetilde{E}_z}{\partial x} + \omega\mu \frac{\partial \widetilde{H}_z}{\partial y}\right), \tag{8.89a}$$

$$\widetilde{E}_y = \frac{j}{k_c^2}\left(-\beta \frac{\partial \widetilde{E}_z}{\partial y} + \omega\mu \frac{\partial \widetilde{H}_z}{\partial x}\right), \tag{8.89b}$$

$$\widetilde{H}_x = \frac{j}{k_c^2}\left(\omega\varepsilon \frac{\partial \widetilde{E}_z}{\partial y} - \beta \frac{\partial \widetilde{H}_z}{\partial x}\right), \tag{8.89c}$$

$$\widetilde{H}_y = \frac{-j}{k_c^2}\left(\omega\varepsilon \frac{\partial \widetilde{E}_z}{\partial x} + \beta \frac{\partial \widetilde{H}_z}{\partial y}\right). \tag{8.89d}$$

Here

$$k_c^2 = k^2 - \beta^2 = \omega^2\mu\varepsilon - \beta^2, \tag{8.90}$$

where k is the ***unbounded-medium wavenumber*** earlier defined as

$$k = \omega\sqrt{\mu\varepsilon}. \tag{8.91}$$

For reasons that become clear later (in Section 8-8), the constant k_c is called the ***cutoff wavenumber***. In view of Eqs. (8.89a-d), the x and y components of $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$ now can be found readily—so long as we have mathematical expressions for $\widetilde{E}_z$ and $\widetilde{H}_z$. For the TE mode, $\widetilde{E}_z = 0$, so all we need to know is $\widetilde{H}_z$, and the converse is true for the TM case.

8-8 TM Modes in Rectangular Waveguide

In the preceding section, we developed expressions for $\widetilde{E}_x$, $\widetilde{E}_y$, $\widetilde{H}_x$, and $\widetilde{H}_y$ in terms of $\widetilde{E}_z$ and $\widetilde{H}_z$. Since $\widetilde{H}_z = 0$ *for the TM mode*, our task reduces to obtaining a valid solution for $\widetilde{E}_z$. Our starting point is the ***homogeneous wave equation*** for $\widetilde{\mathbf{E}}$. For a lossless medium characterized by an unbounded-medium wavenumber k, the wave equation is given by Eq. (7.19) as

$$\nabla^2\widetilde{\mathbf{E}} + k^2\widetilde{\mathbf{E}} = 0. \tag{8.92}$$

To satisfy Eq. (8.92), each of its $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, and $\hat{\mathbf{z}}$ components has to be satisfied independently. Its $\hat{\mathbf{z}}$ component is given by

$$\frac{\partial^2 \widetilde{E}_z}{\partial x^2} + \frac{\partial^2 \widetilde{E}_z}{\partial y^2} + \frac{\partial^2 \widetilde{E}_z}{\partial z^2} + k^2 \widetilde{E}_z = 0. \tag{8.93}$$

By adopting the mathematical form given by Eq. (8.85), namely,

$$\widetilde{E}_z(x,y,z) = \widetilde{e}_z(x,y)\, e^{-j\beta z}, \tag{8.94}$$

Eq. (8.93) reduces to

$$\frac{\partial^2 \widetilde{e}_z}{\partial x^2} + \frac{\partial^2 \widetilde{e}_z}{\partial y^2} + k_c^2 \widetilde{e}_z = 0, \tag{8.95}$$

where k_c^2 is as defined by Eq. (8.90).

The form of the partial differential equation (separate, uncoupled derivatives with respect to x and y) allows us to assume a product solution of the form

$$\widetilde{e}_z(x,y) = X(x)\, Y(y). \tag{8.96}$$

Substituting Eq. (8.96) into Eq. (8.95) and then dividing all terms by $X(x)\,Y(y)$ leads to:

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + k_c^2 = 0. \tag{8.97}$$

To satisfy Eq. (8.97), each of the first two terms has to equal a constant. Hence, we define separation constants k_x and k_y such that

$$\frac{d^2X}{dx^2} + k_x^2 X = 0, \tag{8.98a}$$

$$\frac{d^2Y}{dy^2} + k_y^2 Y = 0, \tag{8.98b}$$

and

$$k_c^2 = k_x^2 + k_y^2. \tag{8.99}$$

三个独立的方程 - 每个单位向量 x̂、ŷ 和 ẑ，我们得到以下关系：

$$\frac{\partial \widetilde{e}_z}{\partial y} + j\beta \widetilde{e}_y = -j\omega\mu \widetilde{h}_x, \tag{8.88a}$$

$$-j\beta \widetilde{e}_x - \frac{\partial \widetilde{e}_z}{\partial x} = -j\omega\mu \widetilde{h}_y, \tag{8.88b}$$

$$\frac{\partial \widetilde{e}_y}{\partial x} - \frac{\partial \widetilde{e}_x}{\partial y} = -j\omega\mu \widetilde{h}_z, \tag{8.88c}$$

$$\frac{\partial \widetilde{h}_z}{\partial y} + j\beta \widetilde{h}_y = j\omega\varepsilon \widetilde{e}_x, \tag{8.88d}$$

$$-j\beta \widetilde{h}_x - \frac{\partial \widetilde{h}_z}{\partial x} = j\omega\varepsilon \widetilde{e}_y, \tag{8.88e}$$

$$\frac{\partial \widetilde{h}_y}{\partial x} - \frac{\partial \widetilde{h}_x}{\partial y} = j\omega\varepsilon \widetilde{e}_z. \tag{8.88f}$$

方程 (8.88a-f) 包含了这样一个事实：对 z 的微分相当于乘以 − jβ。通过对这些方程进行代数运算，我们可以得到 ~E 和 ~H 的 x 和 y 分量以其 z 分量的表达式，即

$$\widetilde{E}_x = \frac{-j}{k_c^2}\left(\beta \frac{\partial \widetilde{E}_z}{\partial x} + \omega\mu \frac{\partial \widetilde{H}_z}{\partial y}\right), \tag{8.89a}$$

$$\widetilde{E}_y = \frac{j}{k_c^2}\left(-\beta \frac{\partial \widetilde{E}_z}{\partial y} + \omega\mu \frac{\partial \widetilde{H}_z}{\partial x}\right), \tag{8.89b}$$

$$\widetilde{H}_x = \frac{j}{k_c^2}\left(\omega\varepsilon \frac{\partial \widetilde{E}_z}{\partial y} - \beta \frac{\partial \widetilde{H}_z}{\partial x}\right), \tag{8.89c}$$

$$\widetilde{H}_y = \frac{-j}{k_c^2}\left(\omega\varepsilon \frac{\partial \widetilde{E}_z}{\partial x} + \beta \frac{\partial \widetilde{H}_z}{\partial y}\right). \tag{8.89d}$$

这里

$$k_c^2 = k^2 - \beta^2 = \omega^2\mu\varepsilon - \beta^2, \tag{8.90}$$

其中 k 是无界介质波数，之前定义为

$$k = \omega\sqrt{\mu\varepsilon}. \tag{8.91}$$

由于稍后会变得清楚的原因（在第 8-8 节中），常数 k_c 称为截止波数。鉴于方程式。(8.89a-d)，现在可以很容易地找到 ~E 和 ~H 的 x 和 y 分量——只要我们有 $\widetilde{E}_z$ 和 $\widetilde{H}_z$ 的数学表达式。对于TE模式，$\widetilde{E}_z = 0$，所以我们只需要知道 $\widetilde{H}_z$，对于TM情况则相反。

矩形波导中的 8-8 TM 模式

在上一节中，我们根据 $\widetilde{E}_z$ 和 $\widetilde{H}_z$ 开发了 $\widetilde{E}_x$、$\widetilde{E}_y$、$\widetilde{H}_x$ 和 $\widetilde{H}_y$ 的表达式。由于 TM 模式的 $\widetilde{H}_z = 0$，我们的任务简化为获取 $\widetilde{E}_z$ 的有效解。我们的起点是 ~E 的齐次波动方程。对于以无界介质波数 k 为特征的无损介质，波动方程由式（1）给出。(7.19) 作为

$$\nabla^2\widetilde{\mathbf{E}} + k^2\widetilde{\mathbf{E}} = 0. \tag{8.92}$$

满足方程。(8.92) 中，其每个 x̂、ŷ 和 ẑ 分量都必须独立满足。它的 ẑ 分量由下式给出

$$\frac{\partial^2 \widetilde{E}_z}{\partial x^2} + \frac{\partial^2 \widetilde{E}_z}{\partial y^2} + \frac{\partial^2 \widetilde{E}_z}{\partial z^2} + k^2 \widetilde{E}_z = 0. \tag{8.93}$$

采用方程给出的数学形式。（8.85），即

$$\widetilde{E}_z(x,y,z) = \widetilde{e}_z(x,y)\, e^{-j\beta z}, \tag{8.94}$$

等式。(8.93) 减少为

$$\frac{\partial^2 \widetilde{e}_z}{\partial x^2} + \frac{\partial^2 \widetilde{e}_z}{\partial y^2} + k_c^2 \widetilde{e}_z = 0, \tag{8.95}$$

其中 k_c^2 由方程式定义。（8.90）。

偏微分方程的形式（关于 x 和 y 的独立、非耦合导数）允许我们假设以下形式的乘积解

$$\widetilde{e}_z(x,y) = X(x)\, Y(y). \tag{8.96}$$

代入方程式(8.96) 代入方程。(8.95) 然后将所有项除以 X(x) Y (y) 得出：

$$\frac{1}{X}\frac{d^2X}{dx^2} + \frac{1}{Y}\frac{d^2Y}{dy^2} + k_c^2 = 0. \tag{8.97}$$

满足方程。(8.97)，前两项必须等于一个常数。因此，我们定义分离常数 k_x 和 k_y 使得

$$\frac{d^2X}{dx^2} + k_x^2 X = 0, \tag{8.98a}$$

$$\frac{d^2Y}{dy^2} + k_y^2 Y = 0, \tag{8.98b}$$

和

$$k_c^2 = k_x^2 + k_y^2. \tag{8.99}$$

Before proposing solutions for Eqs. (8.98a and b), we should consider the constraints that the solutions must meet. The electric field $\tilde{E}_z$ is parallel to all four walls of the waveguide. Since $\mathbf{E} = 0$ in the conducting walls, *the boundary conditions require $\tilde{E}_z$ in the waveguide cavity to go to zero as x approaches 0 and a and as y approaches 0 and b* (**Fig. 8-22**). To satisfy these boundary conditions, sinusoidal solutions are chosen for $X(x)$ and $Y(y)$ as:

$$\tilde{e}_z = X(x)\,Y(y) = (A\cos k_x x + B\sin k_x x)(C\cos k_y y + D\sin k_y y). \tag{8.100}$$

These forms for $X(x)$ and $Y(y)$ definitely satisfy the differential equations given by Eqs. (8.98a and b). The boundary conditions for $\tilde{e}_z$ are

$$\tilde{e}_z = 0 \qquad \text{at } x = 0 \text{ and } a, \tag{8.101a}$$

$$\tilde{e}_z = 0 \qquad \text{at } y = 0 \text{ and } b. \tag{8.101b}$$

Satisfying $\tilde{e}_z = 0$ at $x = 0$ requires that we set $A = 0$, and similarly, satisfying $\tilde{e}_z = 0$ at $y = 0$ requires $C = 0$. Satisfying $\tilde{e}_z = 0$ at $x = a$ requires

$$k_x = \frac{m\pi}{a}, \qquad m = 1,2,3,\ldots \tag{8.102a}$$

and similarly, satisfying $\tilde{e}_z = 0$ at $y = b$ requires

$$k_y = \frac{n\pi}{b}, \qquad n = 1,2,3,\ldots \tag{8.102b}$$

Consequently,

$$\tilde{E}_z = \tilde{e}_z e^{-j\beta z} = E_0 \sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \tag{8.103}$$

where $E_0 = BD$ is the amplitude of the wave in the guide. Keeping in mind that $\tilde{H}_z = 0$ for the TM mode, the transverse components of $\tilde{\mathbf{E}}$ and $\tilde{\mathbf{H}}$ now can be obtained by applying Eq. (8.103) to Eqs. (8.89a–d),

$$\tilde{E}_x = \frac{-j\beta}{k_c^2}\left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \tag{8.104a}$$

$$\tilde{E}_y = \frac{-j\beta}{k_c^2}\left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \tag{8.104b}$$

$$\tilde{H}_x = \frac{j\omega\varepsilon}{k_c^2}\left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \tag{8.104c}$$

$$\tilde{H}_y = \frac{-j\omega\varepsilon}{k_c^2}\left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}. \tag{8.104d}$$

Each combination of the integers m and n represents a viable solution—or a mode—denoted TM_{mn}. Associated with each mn mode are specific field distributions for the region inside the guide. **Figure 8-23** depicts the **E** and **H** field lines for the TM_{11} mode across two different cross sections of the guide.

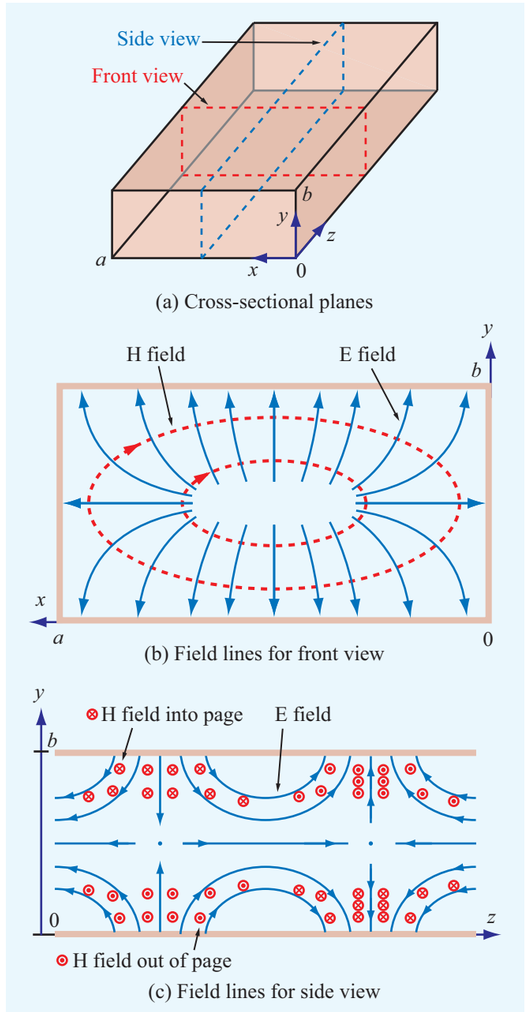


Figure 8-23 TM_{11} electric and magnetic field lines across two cross-sectional planes.

According to Eqs. (8.103) and (8.104e), a rectangular waveguide with cross section $(a \times b)$ can support the propagation of waves with many different, but discrete, field configurations specified by the integers m and n. The only quantity in the

在提出方程式的解决方案之前。（8.98a 和 b），我们应该考虑解必须满足的约束。电场 ~E_z 平行于波导的所有四个壁。由于导电壁中的 E = 0，边界条件要求当 x 接近 0 和 a 以及 y 接近 0 和 b 时，波导腔中的 ~E_z 为零（图 8-22）。为了满足这些边界条件，为 X(x) 和 Y (y) 选择正弦解，如下所示：

$$\tilde{e}_z = X(x)\,Y(y) = (A\cos k_x x + B\sin k_x x)(C\cos k_y y + D\sin k_y y). \tag{8.100}$$

X(x) 和 Y(y) 的这些形式绝对满足方程 1 给出的微分方程。（8.98a 和 b）。~e_z 的边界条件是

$$\tilde{e}_z = 0 \qquad \text{at } x = 0 \text{ and } a, \tag{8.101a}$$

$$\tilde{e}_z = 0 \qquad \text{at } y = 0 \text{ and } b. \tag{8.101b}$$

在 x = 0 处满足 ~$e_z = 0$ 需要我们设置 A = 0，同样，在 y = 0 处满足 ~$e_z = 0$ 需要 C = 0。在 x = a 处满足 ~$e_z = 0$ 需要

$$k_x = \frac{m\pi}{a}, \qquad m = 1,2,3,\ldots \tag{8.102a}$$

同样，在 y = b 处满足 ~$e_z = 0$ 需要

$$k_y = \frac{n\pi}{b}, \qquad n = 1,2,3,\ldots \tag{8.102b}$$

最后，

$$\tilde{E}_z = \tilde{e}_z e^{-j\beta z} = E_0 \sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \tag{8.103}$$

其中 $E_0 = BD$ 是波导中波的振幅。请记住，对于 TM 模式，~$H_z = 0$，~E 和 ~H 的横向分量现在可以通过应用等式获得。(8.103) 至等式。（8.89a-d），

$$\tilde{E}_x = \frac{-j\beta}{k_c^2}\left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \tag{8.104a}$$

$$\tilde{E}_y = \frac{-j\beta}{k_c^2}\left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \tag{8.104b}$$

$$\tilde{H}_x = \frac{j\omega\varepsilon}{k_c^2}\left(\frac{n\pi}{b}\right) E_0 \sin\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right) e^{-j\beta z}, \tag{8.104c}$$

$$\tilde{H}_y = \frac{-j\omega\varepsilon}{k_c^2}\left(\frac{m\pi}{a}\right) E_0 \cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right) e^{-j\beta z}. \tag{8.104d}$$

整数 m 和 n 的每个组合都代表一个可行的解决方案或模式，表示为 TM_{mn}。与每个 mn 模式相关的是指南内区域的特定场分布。图 8-23 描绘了 TM_{11} 模式在波导两个不同横截面上的 E 和 H 场线。

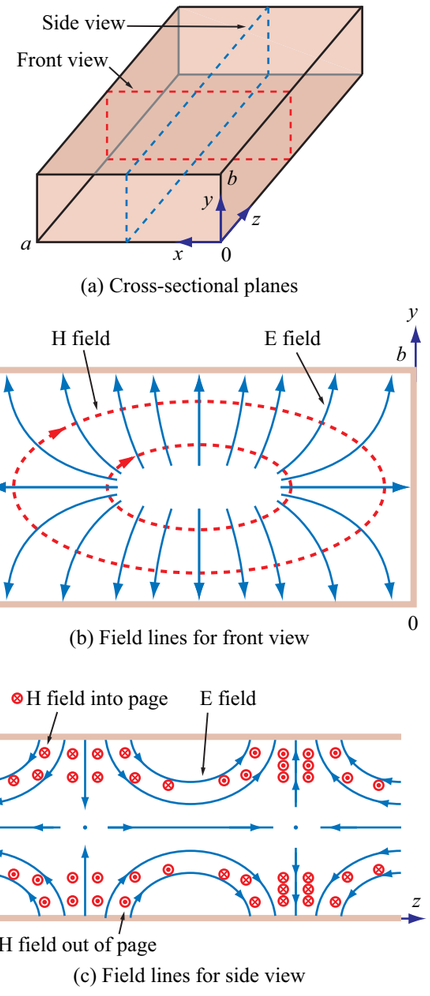


图 8-23 穿过两个横截面的 TM_{11} 电场线和磁场线。

根据方程式。根据（8.103）和（8.104e），具有横截面（a × b）的矩形波导可以支持具有由整数m和n指定的许多不同但离散的场配置的波的传播。中唯一的数量

fields' expressions that we have yet to determine is the propagation constant β, which is contained in the exponential $e^{-j\beta z}$. By combining Eqs. (8.90), (8.99), and (8.102), we obtain the following expression for β:

$$\beta = \sqrt{k^2 - k_c^2} = \sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}. \quad (8.105)$$

(TE and TM)

Even though the expression for β was derived for TM modes, it is equally applicable to TE modes.

The exponential $e^{-j\beta z}$ describes a wave traveling in the $+z$ direction—provided that β is real—which corresponds to $k > k_c$. If $k < k_c$, β becomes imaginary: $\beta = -j\alpha$ with α real, in which case $e^{-j\beta z} = e^{-\alpha z}$, yielding ***evanescent waves*** characterized by amplitudes that decay rapidly with z due to the attenuation function $e^{-\alpha z}$. Corresponding to each mode (m,n), there is a ***cutoff frequency*** f_{mn} at which $\beta = 0$. By setting $\beta = 0$ in Eq. (8.105) and then solving for f, we have

$$f_{mn} = \frac{u_{p_0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \quad (8.106)$$

(TE and TM)

where $u_{p_0} = 1/\sqrt{\mu\varepsilon}$ is the phase velocity of a TEM wave in an unbounded medium with constitutive parameters ε and μ.

► A wave in a given mode can propagate through the guide only if its frequency $f > f_{mn}$, as only then β = real. ◄

The mode with the lowest cutoff frequency is known as the ***dominant mode***. *The dominant mode is TM_{11} among TM modes and TE_{10} among TE modes. Whereas a value of zero for m or n is allowed for TE modes, it is not for TM modes* [because if either m or n is zero, $\widetilde{E}_z$ in Eq. (8.103) becomes zero and all other field components vanish as well].

By combining Eqs. (8.105) and (8.106), we can express β in terms of f_{mn},

$$\beta = \frac{\omega}{u_{p_0}}\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}. \quad \textbf{(TE and TM)} \quad (8.107)$$

The phase velocity of a TE or TM wave in a waveguide is

$$u_p = \frac{\omega}{\beta} = \frac{u_{p_0}}{\sqrt{1 - (f_{mn}/f)^2}}. \quad \textbf{(TE and TM)} \quad (8.108)$$

The transverse electric field consists of components $\widetilde{E}_x$ and $\widetilde{E}_y$ given by Eqs. (8.104a and b). For a wave traveling in the $+z$ direction, the magnetic field associated with $\widetilde{E}_x$ is $\widetilde{H}_y$ [according to the right hand rule given by Eq. (7.39a)]. Similarly, the magnetic field associated with $\widetilde{E}_y$ is $-\widetilde{H}_x$. The ratios, obtained by employing Eq. (8.104e), constitute the wave impedance in the guide,

$$Z_{TM} = \frac{\widetilde{E}_x}{\widetilde{H}_y} = -\frac{\widetilde{E}_y}{\widetilde{H}_x} = \frac{\beta\eta}{k} = \eta\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}, \quad (8.109)$$

where $\eta = \sqrt{\mu/\varepsilon}$ is the intrinsic impedance of the dielectric material filling the guide.

Example 8-8: Mode Properties

A TM wave propagating in a dielectric-filled waveguide of unknown permittivity has a magnetic field with the y component given by

$$H_y = 6\cos(25\pi x)\sin(100\pi y) \times \sin(1.5\pi \times 10^{10}t - 109\pi z) \quad \text{(mA/m)}.$$

If the guide dimensions are $a = 2b = 4$ cm, determine (a) the mode numbers, (b) the relative permittivity of the material in the guide, and (c) the phase velocity, and (d) obtain an expression for E_x.

Solution: (a) By comparison with the expression for $\widetilde{H}_y$ given by Eq. (8.104d), we deduce that the argument of x is $(m\pi/a)$ and the argument of y is $(n\pi/b)$. Hence,

$$25\pi = \frac{m\pi}{4\times10^{-2}}, \qquad 100\pi = \frac{n\pi}{2\times10^{-2}},$$

which yield $m = 1$ and $n = 2$. Therefore, the mode is TM_{12}.

(b) The second sine function in the expression for H_y represents $\sin(\omega t - \beta z)$, which means that

$$\omega = 1.5\pi \times 10^{10} \text{ (rad/s) or } f = 7.5 \text{ GHz},$$
$$\beta = 109\pi \text{ (rad/m)}.$$

By rewriting Eq. (8.105) to obtain an expression for $\varepsilon_r = \varepsilon/\varepsilon_0$ in terms of the other quantities, we have

$$\varepsilon_r = \frac{c^2}{\omega^2}\left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right],$$

场表达式中尚待确定的是传播常数 β，它出现在指数因子 $e^{-j\beta z}$ 中。联立式（8.90）、（8.99）和（8.102），可得 β 的表达式：

$$\beta = \sqrt{k^2 - k_c^2} = \sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}. \quad (8.105)$$

(TE 和 TM)

尽管 β 的表达式是针对 TM 模式导出的，但它同样适用于 TE 模式。

指数 $e^{-j\beta z}$ 描述了沿 +z 方向传播的波（假设 β 为实数），对应于 $k > k_c$。如果 $k < k_c$，β 变为虚数：$\beta = -j\alpha$ 且 α 为实数，在这种情况下 $e^{-j\beta z} = e^{-\alpha z}$，产生倏逝波，其特征是由于衰减函数 $e^{-\alpha z}$，其振幅随 z 快速衰减。对应于每个模态(m，n)，存在一个截止频率 f_{mn}，其中 β =为 0。通过将式(1)中的 β =设置为0，可以得到：(8.105) 然后求解 f，我们有

$$f_{mn} = \frac{u_{p_0}}{2}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}, \quad (8.106)$$

(TE and TM)

其中 $u_{p_0} = 1/\sqrt{\mu\varepsilon}$ 是具有本构参数 ε 和 μ 的无界介质中 TEM 波的相速度。

► 给定模式下的波只有其频率为 $f > f_{mn}$ 时才能通过波导传播，只有 β = 才是真实的。◄

具有最低截止频率的模式称为主模式。TM模式中的主导模式是TM_{11}，TE模式中的主导模式是TE_{10}。虽然 TE 模式允许 m 或 n 为零，但 TM 模式则不允许 [因为如果 m 或 n 为零，则方程 1 中的 $\widetilde{E}_z$ 为 0。（8.103）变为零，所有其他场分量也消失]。

通过结合等式。(8.105) 和 (8.106)，我们可以用 f_{mn} 来表示 β，

$$\beta = \frac{\omega}{u_{p_0}}\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}. \quad \textbf{(TE and TM)} \quad (8.107)$$

波导中 TE 或 TM 波的相速度为

$$u_p = \frac{\omega}{\beta} = \frac{u_{p_0}}{\sqrt{1 - (f_{mn}/f)^2}}. \quad \textbf{(TE and TM)} \quad (8.108)$$

横向电场由方程 1 给出的分量 $\widetilde{E}_x$ 和 $\widetilde{E}_y$ 组成。（8.104a 和 b）。对于沿 +z 方向传播的波，与 $\widetilde{E}_x$ 相关的磁场为 $\widetilde{H}_y$ [根据等式 1 给出的右手法则] (7.39a)]。类似地，与 $\widetilde{E}_y$ 相关的磁场是 $-\widetilde{H}_x$。通过使用等式获得的比率。(8.104e)，构成波导中的波阻抗，

$$Z_{TM} = \frac{\widetilde{E}_x}{\widetilde{H}_y} = -\frac{\widetilde{E}_y}{\widetilde{H}_x} = \frac{\beta\eta}{k} = \eta\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}, \quad (8.109)$$

其中 $\eta = \sqrt{\mu/\varepsilon}$ 是填充波导的介电材料的固有阻抗。

例 8-8：模式属性

TM 波在介电常数未知的电介质填充波导中传播，其磁场的 y 分量为

$$H_y = 6\cos(25\pi x)\sin(100\pi y) \times \sin(1.5\pi \times 10^{10}t - 109\pi z) \quad \text{(mA/m)}.$$

如果波导尺寸为 a = 2b = 4 cm，则确定 (a) 模数、(b) 波导中材料的相对介电常数、(c) 相速度，以及 (d) 获得 E_x 的表达式。

解：(a) 通过与式(1)给出的 $\widetilde{H}_y$ 的表达式进行比较。(8.104d)，我们推导出 x 的参数是 (mπ/a)，y 的参数是 (nπ/b)。因此，

$$25\pi = \frac{m\pi}{4\times10^{-2}}, \qquad 100\pi = \frac{n\pi}{2\times10^{-2}},$$

产生 m = 1 和 n = 2。因此，众数为 TM_{12}。

(b) H_y 表达式中的第二个正弦函数表示 sin(ω t − β z)，这意味着

$$\omega = 1.5\pi \times 10^{10} \text{ (rad/s) or } f = 7.5 \text{ GHz},$$
$$\beta = 109\pi \text{ (rad/m)}.$$

通过重写方程。(8.105) 为了获得 $\varepsilon_r = \varepsilon/\varepsilon_0$ 用其他量表示的表达式，我们有

$$\varepsilon_r = \frac{c^2}{\omega^2}\left[\beta^2 + \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right],$$

where c is the speed of light. Inserting the available values, we obtain

$$\varepsilon_r = \frac{(3\times10^8)^2}{(1.5\pi\times10^{10})^2} \cdot \left[(109\pi)^2 + \left(\frac{\pi}{4\times10^{-2}}\right)^2 + \left(\frac{2\pi}{2\times10^{-2}}\right)^2\right] = 9.$$

(c)

$$u_p = \frac{\omega}{\beta} = \frac{1.5\pi\times10^{10}}{109\pi} = 1.38\times10^8 \text{ m/s},$$

which is slower than the speed of light. However, as explained later in Section 8-10, the phase velocity in a waveguide may exceed c, but the velocity with which energy is carried down the guide is the group velocity u_g, which is never greater than c.

(d) From Eq. (8.109),

$$Z_{TM} = \eta\sqrt{1-(f_{12}/f)^2}.$$

Application of Eq. (8.106) yields $f_{12} = 5.15$ GHz for the TM_{12} mode. Using that in the expression for Z_{TM}, in addition to $f = 7.5$ GHz and

$$\eta = \sqrt{\mu/\varepsilon} = \frac{\sqrt{\mu_0/\varepsilon_0}}{\sqrt{\varepsilon_r}} = \frac{377}{\sqrt{9}} = 125.67\ \Omega,$$

gives

$$Z_{TM} = 91.3\ \Omega.$$

Hence,

$$\begin{aligned} E_x = Z_{TM}H_y &= 91.3\times6\cos(25\pi x)\sin(100\pi y) \\ &\quad\times\sin(1.5\pi\times10^{10}t - 109\pi z) \qquad \text{(mV/m)} \\ &= 0.55\cos(25\pi x)\sin(100\pi y) \\ &\quad\times\sin(1.5\pi\times10^{10}t - 109\pi z) \qquad \text{(V/m)}. \end{aligned}$$

Concept Question 8-8: What are the primary limitations of coaxial cables at frequencies higher than 30 GHz?

Concept Question 8-9: Can a TE mode have a zero magnetic field along the direction of propagation?

Concept Question 8-10: What is the rationale for choosing a solution for $\widetilde{e}_z$ that involves sine and cosine functions?

Concept Question 8-11: What is an evanescent wave?

Exercise 8-10: For a square waveguide with $a = b$, what is the value of the ratio $\widetilde{E}_x/\widetilde{E}_y$ for the TM_{11} mode?

Answer: $\tan(\pi y/a)/\tan(\pi x/a)$.

Exercise 8-11: What is the cutoff frequency for the dominant TM mode in a waveguide filled with a material with $\varepsilon_r = 4$? The waveguide dimensions are $a = 2b = 5$ cm.

Answer: For TM_{11}, $f_{11} = 3.35$ GHz.

Exercise 8-12: What is the magnitude of the phase velocity of a TE or TM mode at $f = f_{mn}$?

Answer: $u_p = \infty$! [See explanation in Section 8-10.]

8-9 TE Modes in Rectangular Waveguide

In the TM case, where the wave has no magnetic field component along the z direction (i.e., $\widetilde{H}_z = 0$), we started our treatment in the preceding section by obtaining a solution for $\widetilde{E}_z$, and then we used it to derive expressions for the tangential components of $\widetilde{\mathbf{E}}$ and $\widetilde{\mathbf{H}}$. For the TE case, the same basic procedure can be applied, except we reversed the roles of $\widetilde{E}_z$ and $\widetilde{H}_z$. Such a process leads to:

$$\widetilde{E}_x = \frac{j\omega\mu}{k_c^2}\left(\frac{n\pi}{b}\right)H_0\cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)e^{-j\beta z}, \quad (8.110a)$$

$$\widetilde{E}_y = \frac{-j\omega\mu}{k_c^2}\left(\frac{m\pi}{a}\right)H_0\sin\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-j\beta z}, \quad (8.110b)$$

$$\widetilde{H}_x = \frac{j\beta}{k_c^2}\left(\frac{m\pi}{a}\right)H_0\sin\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-j\beta z}, \quad (8.110c)$$

$$\widetilde{H}_y = \frac{j\beta}{k_c^2}\left(\frac{n\pi}{b}\right)H_0\cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)e^{-j\beta z}, \quad (8.110d)$$

$$\widetilde{H}_z = H_0\cos\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-j\beta z}, \quad (8.110e)$$

and of course, $\widetilde{E}_z = 0$. The expressions for f_{mn}, β, and u_p given earlier by Eqs. (8.106), (8.107), and (8.108) remain unchanged.

► Because not all of the fields vanish if m or n assume a value of zero, the lowest order TE mode is TE_{10} if $a > b$ or TE_{01} if $a < b$. It is customary to assign a to be the longer dimension, in which case the TE_{10} mode is the de facto dominant mode. ◄

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其中c是光速。插入可用值，我们得到

$$\varepsilon_r = \frac{(3\times10^8)^2}{(1.5\pi\times10^{10})^2} \cdot \left[(109\pi)^2 + \left(\frac{\pi}{4\times10^{-2}}\right)^2 + \left(\frac{2\pi}{2\times10^{-2}}\right)^2\right] = 9.$$

(c)

$$u_p = \frac{\omega}{\beta} = \frac{1.5\pi\times10^{10}}{109\pi} = 1.38\times10^8 \text{ m/s},$$

它比光速慢。然而，正如稍后第 8-10 节中所解释的，波导中的相速度可能超过 c，但能量沿着波导传递的速度是群速度 u_g，它永远不会大于 c。

(d) 从方程式(8.109),

$$Z_{TM} = \eta\sqrt{1-(f_{12}/f)^2}.$$

方程的应用(8.106) 为 TM_{12} 模式产生 $f_{12} = 5.15$ GHz。在 Z_{TM} 的表达式中使用该值，此外还有 f = 7.5 GHz 和

$$\eta = \sqrt{\mu/\varepsilon} = \frac{\sqrt{\mu_0/\varepsilon_0}}{\sqrt{\varepsilon_r}} = \frac{377}{\sqrt{9}} = 125.67\ \Omega,$$

给出

$$Z_{TM} = 91.3\ \Omega.$$

因此，

$$\begin{aligned} E_x = Z_{TM}H_y &= 91.3\times6\cos(25\pi x)\sin(100\pi y) \\ &\quad\times\sin(1.5\pi\times10^{10}t - 109\pi z) \qquad \text{(mV/m)} \\ &= 0.55\cos(25\pi x)\sin(100\pi y) \\ &\quad\times\sin(1.5\pi\times10^{10}t - 109\pi z) \qquad \text{(V/m)}. \end{aligned}$$

概念问题 8-8：频率高于 30 GHz 时同轴电缆的主要限制是什么？

概念问题 8-9：TE 模式沿传播方向的磁场可以为零吗？

概念问题 8-10：选择涉及正弦和余弦函数的 $\widetilde{e}_z$ 解的基本原理是什么？

概念问题 8-11：什么是倏逝波？

Exercise 8-10: For a square waveguide with $a = b$, what is the value of the ratio $\widetilde{E}_x/\widetilde{E}_y$ for the TM_{11} mode?

Answer: $\tan(\pi y/a)/\tan(\pi x/a)$.

Exercise 8-11: What is the cutoff frequency for the dominant TM mode in a waveguide filled with a material with $\varepsilon_r = 4$? The waveguide dimensions are $a = 2b = 5$ cm.

Answer: For TM_{11}, $f_{11} = 3.35$ GHz.

Exercise 8-12: What is the magnitude of the phase velocity of a TE or TM mode at $f = f_{mn}$?

Answer: $u_p = \infty$! [See explanation in Section 8-10.]

8-9 矩形波导中的TE模式

在 TM 情况下，波沿 z 方向没有磁场分量（即 $\widetilde{H}_z = 0$），我们在上一节中通过获得 $\widetilde{E}_z$ 的解开始处理，然后用它推导 $\widetilde{E}$ 和 $\widetilde{H}$ 的切向分量的表达式。对于 TE 情况，可以应用相同的基本过程，只是我们颠倒了 $\widetilde{E}_z$ 和 $\widetilde{H}_z$ 的角色。这样的过程导致：

$$\widetilde{E}_x = \frac{j\omega\mu}{k_c^2}\left(\frac{n\pi}{b}\right)H_0\cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)e^{-j\beta z}, \quad (8.110a)$$

$$\widetilde{E}_y = \frac{-j\omega\mu}{k_c^2}\left(\frac{m\pi}{a}\right)H_0\sin\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-j\beta z}, \quad (8.110b)$$

$$\widetilde{H}_x = \frac{j\beta}{k_c^2}\left(\frac{m\pi}{a}\right)H_0\sin\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-j\beta z}, \quad (8.110c)$$

$$\widetilde{H}_y = \frac{j\beta}{k_c^2}\left(\frac{n\pi}{b}\right)H_0\cos\left(\frac{m\pi x}{a}\right)\sin\left(\frac{n\pi y}{b}\right)e^{-j\beta z}, \quad (8.110d)$$

$$\widetilde{H}_z = H_0\cos\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi y}{b}\right)e^{-j\beta z}, \quad (8.110e)$$

当然，$\widetilde{E}_z = 0$。f_{mn}、β 和 u_p 的表达式前面由等式给出。(8.106)、(8.107)和(8.108)保持不变。

► 因为如果 m 或 n 的值为零，则并非所有场都会消失，因此最低阶 TE 模式是 TE_{10}（如果 a > b）或 TE_{01}（如果 a < b）。通常将 a 指定为较长的维度，在这种情况下，TE_{10} 模式是事实上的主导模式。◄